

SMART INDIA HACKATHON 2026 — SOFTWARE

SIH26003

Ministry of Development of North Eastern Region (MDoNER)

AI-Based Cognitive Gaming & Memory Assistance Platform for Elderly Dementia Patients in NER

A Complete Guide, Technical Blueprint, and Execution Roadmap for building an AI-enabled cognitive assistance platform with adaptive gaming, multilingual support, and offline-first architecture for the North Eastern Region of India.

CATEGORY

Software

THEME

MedTech / HealthTech

DOMAIN

AI / ML / Healthcare

DEADLINE

30 Sept 2026



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Problem Statement — Deep Dive & Analysis

01

Understanding every nuance of SIH26003 to build a winning solution

1.1 Official Problem Statement

SIH26003 — AI-Based Cognitive Gaming & Memory Assistance Platform for Elderly Dementia Patients in NER

Organization:

Ministry of Development of North Eastern Region (MDoNER)

Category:

Software |

Theme:

MedTech / BioTech / HealthTech

1.2 Background (as stated by MDoNER)

The North Eastern Region is witnessing rising cognitive disorders like **dementia** among its elderly population. Remote and rural areas lack access to specialized neurological care, cognitive therapy, and long-term elderly support services due to limited healthcare infrastructure. There is an urgent need for an AI-enabled cognitive gaming solution tailored to dementia patients in NER.

1.3 What the Solution Must Do (Official Requirements)

#	Requirement	Category	Priority
R1	Interactive cognitive games — memory improvement, attention, daily routine recall, pattern recognition	AI/Games	Critical
R2	AI/ML to adapt difficulty levels based on patient performance	AI/ML	Critical
R3	Multilingual and voice-assisted interaction for elderly NER users	Accessibility	Critical
R4	Culturally familiar themes, visuals, sounds, and regional language support	Design	High
R5	Reminders — medicines, hydration, daily activities, medical appointments	Backend	High
R6	Caregiver/healthcare worker monitoring dashboards	Dashboard	High
R7	Low-connectivity operation with offline functionality	Infra	Critical
R8	Accessible via mobile/tablet with elderly-friendly interface	UX	Critical

1.4 Expected Solution Components

 **Adaptive Gaming Engine**

Memory training modules with AI-driven difficulty adaptation that responds to cognitive performance in real time.

 **Voice-Enabled Multilingual UI**

NER language support (Assamese, Bodo, Manipuri, Mizo, Khasi, etc.) with voice commands and TTS.

 **Performance Analytics Dashboard**

Cognitive tracking with trend analysis for caregivers and medical professionals.

 **Smart Reminder System**

Personalized reminders for medication, hydration, exercise, and appointments with gentle nudging.

 **Offline Sync Engine**

 **Secure Patient Data**

Full offline gameplay and data collection with smart sync when connectivity returns.

Encrypted patient records with role-based access for caregivers and healthcare workers.

1.5 Problem Decomposition — What the Judges Want

⚡ Key Insight for SIH Judges

This isn't just a "make some brain games" problem. The judges are from MDoNER — they want to see that you understand the

unique challenges of NER

(linguistic diversity, terrain, connectivity gaps, cultural sensitivity). The winning solution must feel like it was built *for* an elderly person in Tezpur, not just a generic health app with a language toggle.

Dimension	What Judges Evaluate	How to Score High
Problem Understanding	Did you research NER-specific dementia issues?	Cite NER statistics, mention LGBRIMH, reference ARDSI data
Technical Depth	Is the AI/ML component real or just buzzwords?	Explain your RL-based difficulty adaptation with actual algorithm choice
Cultural Sensitivity	Are the games culturally appropriate for NER elderly?	Use NER folk art, festivals, foods, music as game content
Feasibility	Can this actually run in low-connectivity NER areas?	Demonstrate offline-first architecture with sync strategy
Impact	Will this meaningfully help dementia patients?	Show cognitive science backing for each game type

Domain Background — Dementia, NER, & the Opportunity

02

Everything you need to know about the problem domain

2.1 Dementia — Understanding the Disease

Dementia is a **syndrome** (not a single disease) characterized by progressive deterioration of cognitive function — memory, thinking, orientation, comprehension, judgment, and language. It primarily affects people aged 65+, and Alzheimer's disease accounts for 60–70% of cases.

Cognitive Domains Affected:

Domain	What Gets Impaired	Game Type That Helps
Episodic Memory	Recalling recent events, appointments	Story recall, sequence games, photo matching
Working Memory	Holding info while using it	N-back tasks, digit span, card matching
Attention	Focusing, filtering distractions	Visual search, Stroop tasks, sustained attention
Executive Function	Planning, decision-making, problem-solving	Tower puzzles, categorization, sequencing
Visuospatial	Navigation, spatial orientation	Map games, jigsaw puzzles, pattern completion
Language	Word finding, naming, comprehension	Word association, picture naming, storytelling

2.2 The NER Context — Why It Matters

🏞️ Geographic Challenges

8

👴 Elderly Population

~5-8%

States in NER: Assam, Arunachal Pradesh, Manipur, Meghalaya, Mizoram, Nagaland, Tripura, Sikkim. Hilly terrain, poor road connectivity, remote villages.

Estimated dementia prevalence among 60+ population in India. NER data is sparse — itself a problem to highlight.

Linguistic Diversity

200+

Languages/dialects spoken across NER. Major ones: Assamese, Bengali, Bodo, Manipuri (Meitei), Mizo, Khasi, Garo, Nagamese, Nepali.

Connectivity Gaps

Limited

Many rural areas have intermittent 2G/3G at best. Solution MUST work offline with periodic sync.

2.3 Cultural Elements to Incorporate

Making the games feel familiar and warm to elderly NER users is a **core differentiator**. Here are cultural touchpoints to weave into the games:

State	Cultural Elements	Game Integration Ideas
Assam	Bihu dance, Gamosa textile, Rhino, Tea gardens, Barak Valley	Pattern matching with Gamosa weaving patterns; Tea plantation memory game
Manipur	Ras Lila dance, Kangla Fort, Loktak Lake, Polo origin	Sequence recall with Ras Lila dance steps; Loktak Lake ecosystem puzzle
Meghalaya	Living root bridges, Shillong, Nongkrem dance, Khasi Hills	Spatial navigation through living root bridges; rain pattern recognition
Mizoram	Cheraw (Bamboo dance), Mizo festivals, Handloom	Rhythm-based attention game; handloom pattern completion
Nagaland	Hornbill Festival, Naga shawls, Wood carving	Color matching with Naga shawl patterns; festival sequence recall
Tripura	Neer Mahal, Bamboo craft, Garia festival	Architecture memory game; bamboo craft assembly puzzle
Arunachal	Tawang Monastery, Monpa masks, Ziro Valley	Mask matching game; valley ecosystem sorting
Sikkim	Kangchenjunga, Buddhist monasteries, Orchids	Mountain peak identification; orchid classification game

2.4 Existing Solutions & Gap Analysis

Existing Solution	What It Does	Gap / Why It Fails for NER
Lumosity / BrainHQ	Cognitive training games	English only, needs internet, not dementia-specific, no NER culture
Sea Hero Quest	Spatial navigation for dementia research	Data collection tool, not therapeutic. No language support
MindMate	Brain games + nutrition for dementia	Western-centric, English/European languages only
Rendever	VR reminiscence therapy	Requires VR headset — impractical for NER elderly
NeuroNation	Personalized brain training	No offline mode, no NER language support

 Your Differentiator

No existing solution combines: (1) AI-adaptive cognitive games (2) NER cultural context (3) NER language support with voice (4) Offline-first (5) Caregiver dashboards (6) Reminder system — all in one platform. This is your USP.

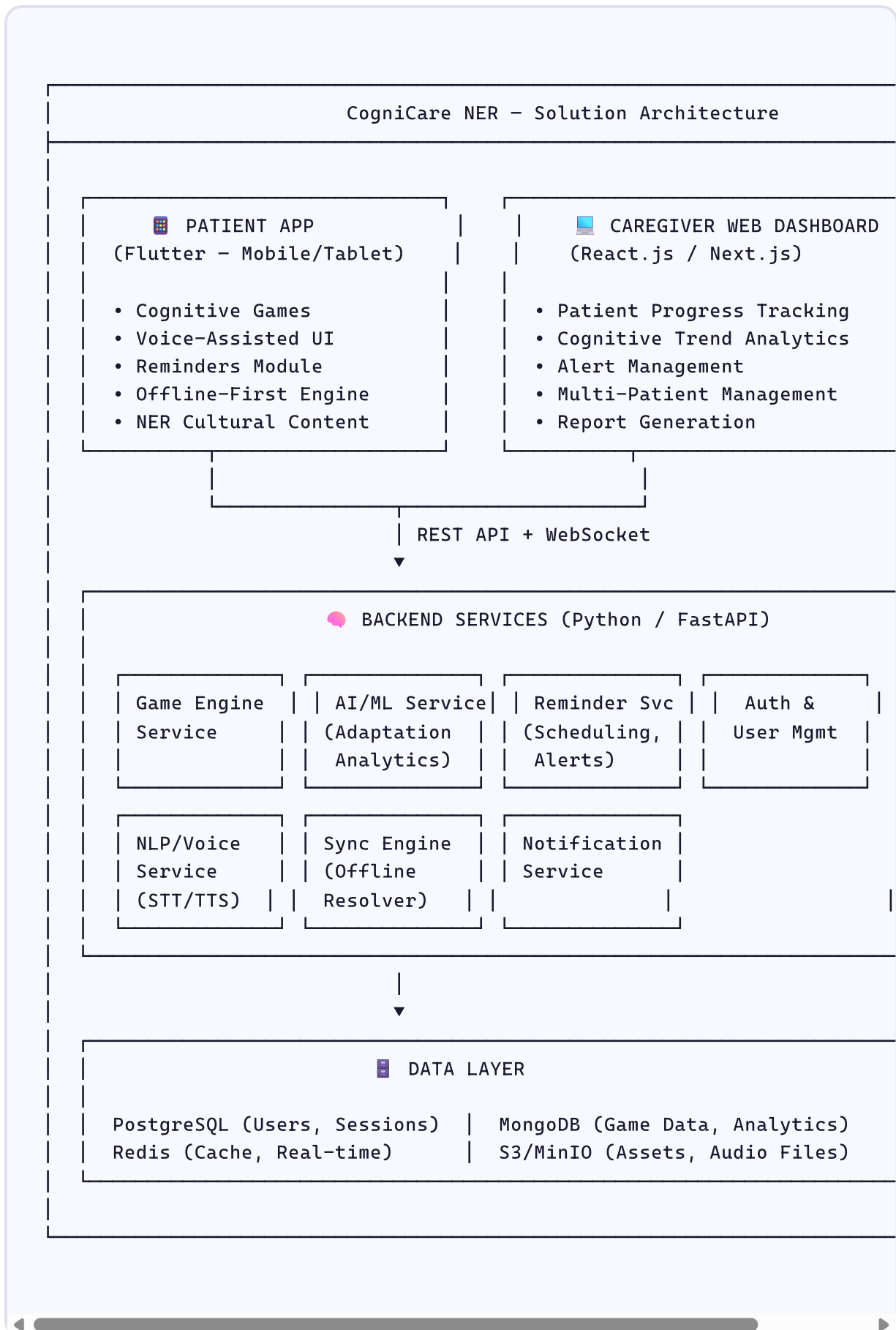
Solution Overview — The "CogniCare NER" Platform

High-level architecture and product vision

3.1 Product Vision

CogniCare NER is an AI-powered mobile platform that provides culturally rooted cognitive games, personalized memory assistance, and caregiver monitoring — designed from the ground up for elderly dementia patients in India's North Eastern Region. It works offline, speaks their language, and adapts to their cognitive level.

3.2 Solution Architecture — Bird's Eye View



3.3 Module Map

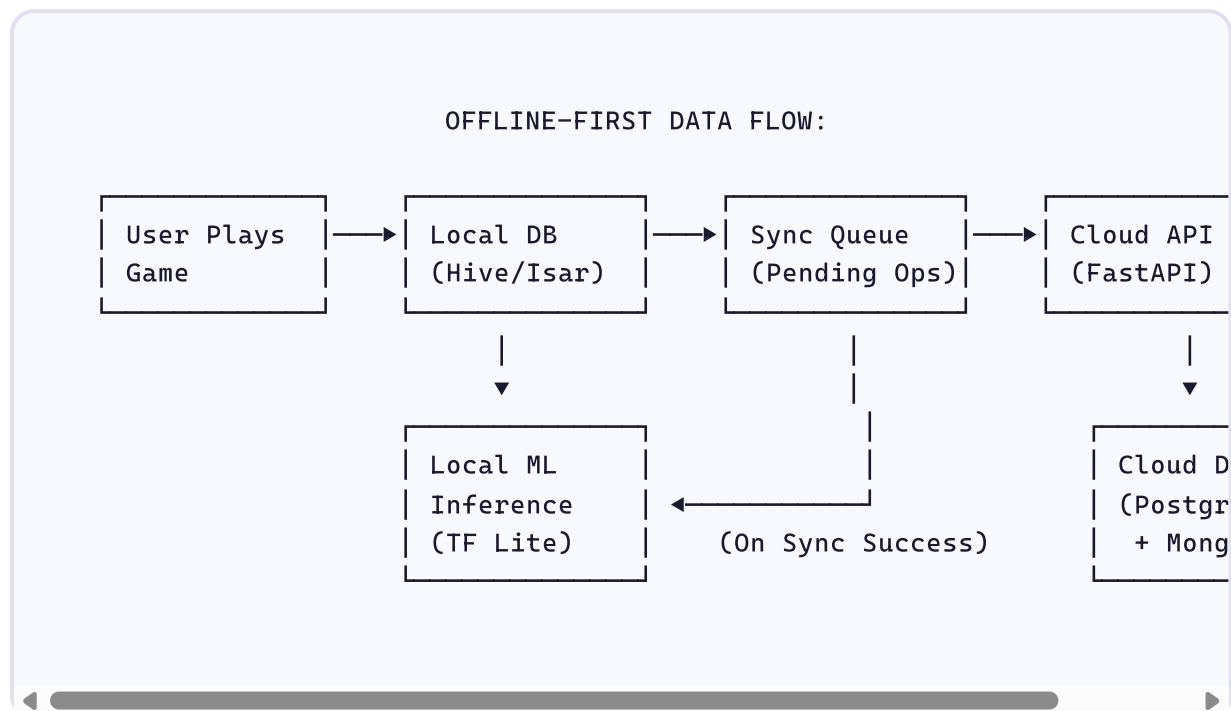
Module	Purpose	Key Technology
 Adaptive Games Engine	6 cognitive game types with RL-based difficulty	Flutter + TensorFlow Lite
 AI Personalization	Tracks cognitive patterns, adapts difficulty, detects decline	Python, scikit-learn, PyTorch
 Voice & Language	STT/TTS in 8+ NER languages, voice commands	Google Cloud Speech / IndicTTS
 Caregiver Dashboard	Analytics, alerts, patient management	React + Chart.js / Recharts
 Reminder System	Medicine, hydration, activity, appointment reminders	Flutter Local Notifications + FCM
 Offline Engine	Local DB, queue-based sync, conflict resolution	Hive/Isar + Custom Sync Protocol
 Security Layer	Encryption, RBAC, audit logs	JWT, AES-256, Role-based access

System Architecture — End-to-End Technical Design

Detailed technical architecture with data flows and component interactions

4.1 Architecture Pattern: Offline-First + Event-Driven

Given NER's connectivity constraints, we adopt an **Offline-First architecture** where the mobile app is the primary data store and the cloud backend is the sync target. All interactions happen locally first, then sync when connectivity is available.



4.2 API Architecture

Endpoint Group	Base Path	Methods	Purpose
Auth	<code>/api/v1/auth</code>	POST	Login, register, refresh token
Patients	<code>/api/v1/patients</code>	CRUD	Patient profile management
Game Sessions	<code>/api/v1/sessions</code>	POST, GET	Submit game results, fetch history
Analytics	<code>/api/v1/analytics</code>	GET	Cognitive trend data, reports
Reminders	<code>/api/v1/reminders</code>	CRUD	Manage reminder schedules
Sync	<code>/api/v1/sync</code>	POST, GET	Bidirectional data sync
Alerts	<code>/api/v1/alerts</code>	GET, PATCH	Caregiver alerts and acknowledgments
Content	<code>/api/v1/content</code>	GET	Download game assets, cultural packs

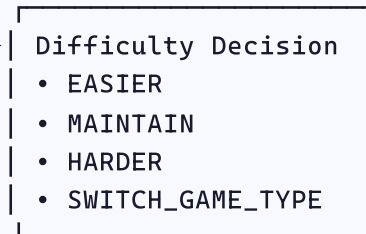
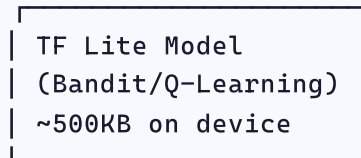
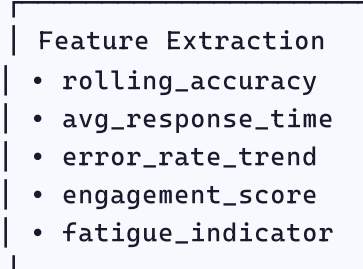
4.3 On-Device ML Architecture

Why On-Device ML?

Since the app must work offline, we run a lightweight TensorFlow Lite model on-device for real-time difficulty adaptation. The full model trains in the cloud and pushes updated weights via sync.

ON-DEVICE ML PIPELINE:

Game Metrics (accuracy, time, errors)



Core Module 1 — Adaptive Cognitive Games Engine

The heart of the platform — 6 game types targeting different cognitive domains

5.1 Game Catalog

Game	Cognitive Domain	Mechanic	NER Cultural Theme Example	Difficulty Params
 Card Match	Working Memory	Flip cards to find matching pairs	Naga shawl patterns, Assamese Gamosa motifs	Grid size (2×2 → 6×6), timeout, decoy cards
 Sequence Recall	Episodic Memory	Remember and reproduce sequences of items/sounds	Bihu dance steps, traditional festival ritual order	Sequence length (3 → 10), display time, interference
 Spot the Difference	Attention	Find differences between two similar NER scenes	Market scenes, festival images, landscape photos	Number of differences, subtlety, time limit
 Pattern Completion	Visuospatial + Executive	Complete missing parts of patterns	Mizo bamboo dance formations, weaving patterns	Pattern complexity, missing elements count
 Word Association	Language	Connect related words, name pictures	Local foods, animals, festival items in NER languages	Category breadth, response time, hint availability
 Daily Routine Sort	Executive Function	Arrange daily activities in correct order	Traditional NER daily routines (tea plucking, weaving, cooking)	Number of steps, distractor items, time pressure

5.2 Difficulty Adaptation Algorithm

Approach: Multi-Armed Bandit with Contextual Features

We use a **Contextual Multi-Armed Bandit (CMAB)** approach — specifically Thompson Sampling — to balance exploration (trying new difficulty levels) with exploitation (staying at the optimal level). This is superior to simple threshold-based adaptation because:

- It handles the **exploration-exploitation tradeoff** — sometimes the best difficulty isn't the one with the highest past score
- It naturally **adapts to changing cognitive states** (good days vs bad days)
- It's **lightweight enough for on-device inference**

```
# Simplified Adaptive Difficulty Algorithm (Python pseudocode)

class CognitiveAdaptiveEngine:
    """Thompson Sampling-based difficulty adaptation."""

    def __init__(self, difficulty_levels=5):
        self.n_levels = difficulty_levels
        # Beta distribution parameters for each level
        self.alpha = np.ones(difficulty_levels) # successes + 1
        self.beta = np.ones(difficulty_levels)  # failures + 1

    def select_difficulty(self, patient_context):
        """Select next difficulty using Thompson Sampling."""
        # Sample from Beta distribution for each level
        samples = [
            np.random.beta(self.alpha[i], self.beta[i])
            for i in range(self.n_levels)
        ]

        # Apply context adjustment (time of day, fatigue, recent trend)
        adjusted = self._apply_context(samples, patient_context)

        # Select level that maximizes expected engagement
        # (not just accuracy - we want the "flow zone")
        return np.argmax(adjusted)

    def update(self, level, performance):
        """Update beliefs after a game session."""
        # "Success" = patient was in the flow zone (60-80% accuracy)
        in_flow = 0.6 ≤ performance.accuracy ≤ 0.8

        if in_flow and performance.engagement > 0.5:
            self.alpha[level] += 1 # reward
        else:
```

```

        self.beta[level] += 1    # penalty

    def _apply_context(self, samples, ctx):
        """Adjust for patient-specific factors."""
        adjustments = np.ones_like(samples)

        if ctx.time_since_last_session > 48: # hours
            adjustments *= 0.8 # ease back in

        if ctx.fatigue_score > 0.7:
            adjustments[3:] *= 0.5 # avoid hard levels

        if ctx.recent_trend == 'declining':
            adjustments *= 0.9 # shift toward easier

        return np.array(samples) * adjustments

```

5.3 Flow Zone Targeting

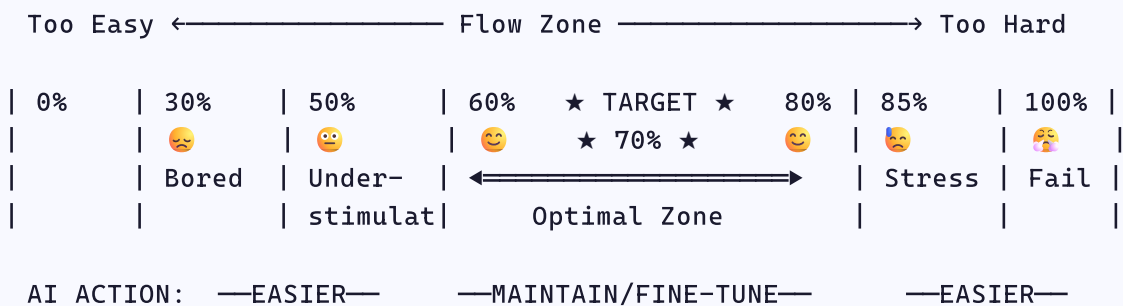
The "Flow Zone" Concept

Research shows cognitive training is most beneficial when the patient is in the **flow zone**

— challenged enough to engage but not so hard that they feel frustrated. Our AI targets **60–80% accuracy**

as the sweet spot. If accuracy drops below 50%, the game gets easier. If it rises above 85%, it gets harder.

FLOW ZONE TARGETING:



Core Module 2 — AI/ML Personalization & Analytics

Tracking cognitive trends, detecting decline, and personalizing the experience

6.1 Data Collection per Game Session

Metric	What It Measures	How It's Used
accuracy	% correct answers	Primary performance indicator
response_time_avg	Average time per interaction (ms)	Processing speed tracking
response_time_variance	Consistency of responses	Attention stability indicator
error_pattern	Types of errors made	Identifies specific cognitive weaknesses
session_duration	How long the patient played	Engagement and fatigue tracking
hint_usage	How often hints were requested	Self-awareness and difficulty calibration
pause_frequency	Number of pauses during gameplay	Fatigue and confusion detection
touch_precision	How precisely they tap targets	Motor function + attention proxy

6.2 Cognitive Trend Analysis

Composite Cognitive Score (CCS)

We compute a weighted **Composite Cognitive Score (0–100)** from game performance across all domains:

$$\text{CCS} = w1 \times \text{Memory_Score} + w2 \times \text{Attention_Score} + w3 \times \text{Executive_Score} + w4 \times \text{Language_Score} + w5 \times \text{Visuospatial_Score}$$

```
Where:
w1 = 0.25 (Memory – most critical for dementia)
w2 = 0.20 (Attention)
w3 = 0.20 (Executive Function)
w4 = 0.15 (Language)
w5 = 0.20 (Visuospatial)

Each domain score = f(accuracy, response_time, consistency, improvement_ra
```

6.3 Decline Detection — Alert Triggers

Alert Type	Trigger Condition	Sent To	Priority
 Rapid Decline	CCS drops >15 points in 7 days	Caregiver + Healthcare Worker	Urgent
 Gradual Decline	CCS downtrend for >3 consecutive weeks	Caregiver	Warning
 Missed Sessions	No game played in 3+ days	Caregiver	Info
 Domain-Specific Drop	Any single domain score drops >20% in 14 days	Caregiver	Warning
 Improvement Detected	CCS increases >10 points over 14 days	Caregiver (positive reinforcement)	Info

6.4 ML Models Used

Model	Purpose	Algorithm	Where It Runs
Difficulty Adapter	Real-time game difficulty adjustment	Thompson Sampling / Contextual Bandit	On-Device (TF Lite)
Trend Predictor	Predict cognitive trajectory (improving/stable/declining)	LSTM / GRU time-series model	Cloud
Anomaly Detector	Detect sudden behavioral changes	Isolation Forest / Autoencoders	Cloud
Fatigue Classifier	Detect when patient is getting tired	Random Forest on session metrics	On-Device
Game Recommender	Suggest which game to play next	Collaborative Filtering + Content-Based	Hybrid (On-Device fallback)

Core Module 3 — Multilingual & Voice-Assisted Interface07

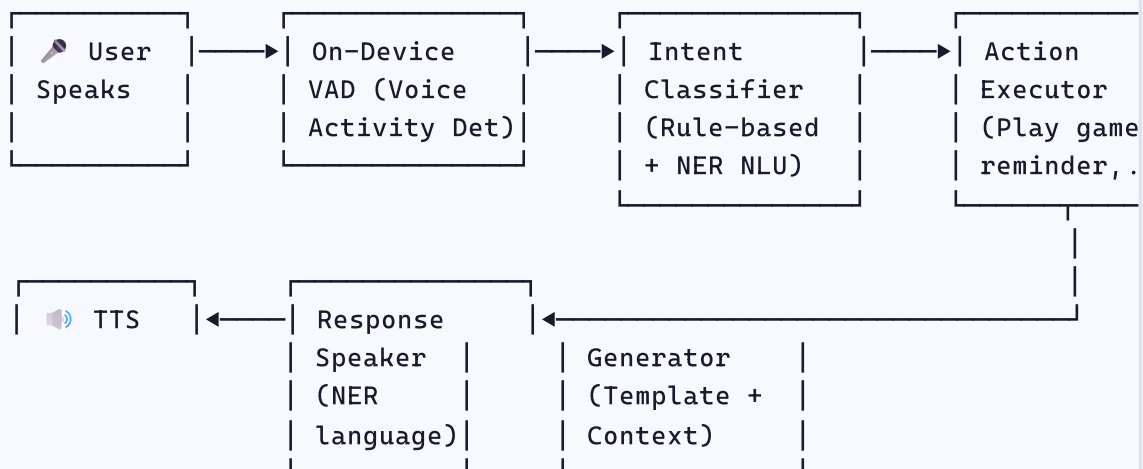
Making the app speak the language of NER's elderly

7.1 Language Support Matrix

Language	Script	State(s)	STT Support	TTS Support	Priority
Assamese	Bengali script	Assam	Google Cloud STT	IndicTTS	P0
Bengali	Bengali script	Tripura, Assam	Google Cloud STT	Google TTS	P0
Hindi	Devanagari	All (lingua franca)	Google Cloud STT	Google TTS	P0
Manipuri (Meitei)	Meetei Mayek	Manipur	Limited — custom model	IndicTTS	P1
Mizo	Latin script	Mizoram	Custom fine-tuned	Custom TTS	P1
Bodo	Devanagari	Assam	Custom fine-tuned	IndicTTS (experimental)	P2
Khasi	Latin script	Meghalaya	Custom fine-tuned	Custom TTS	P2
Nepali	Devanagari	Sikkim	Google Cloud STT	Google TTS	P1

7.2 Voice Interaction Architecture

VOICE INTERACTION FLOW:



VOICE COMMANDS SUPPORTED:

- "Game khelna hai" / "Khel suru karo" → Start a game
 - "Dawai ka time" → Check medicine reminder
- "Aaj kaise khela?" → Today's performance summary
 - "Help" / "Sahayata" → Context-sensitive help
 - "Band karo" → Pause/stop current game

7.3 Offline Language Support

Offline-Capable Voice Processing

For offline operation, we pre-download lightweight voice models to the device:

- **Vosk**
 - Open-source offline STT (supports Hindi, can be fine-tuned for Assamese/Bengali)
- **eSpeakNG + MBROLA**
 - Lightweight offline TTS for multiple Indic languages
- **Rule-based NLU**
 - Simple keyword-matching for voice commands (no heavy NLP needed)

Online mode uses Google Cloud Speech API for higher accuracy when connectivity is available.

Core Module 4 — Caregiver Dashboard & Alert System 08

Giving caregivers and healthcare workers actionable insights

8.1 Dashboard Features

Cognitive Trend View

Interactive line charts showing CCS over time with domain breakdowns. Compare week-over-week. Spot trends before clinical visits.

Session Analytics

Per-session breakdown: games played, scores, time spent, difficulty levels reached. Heatmap of active hours.

Alert Center

Color-coded alerts (urgent/warning/info). One-tap acknowledgment. Push notifications to caregiver's phone.

Patient Profiles

Multi-patient management. Medical history notes. Custom game preferences. Reminder configurations.

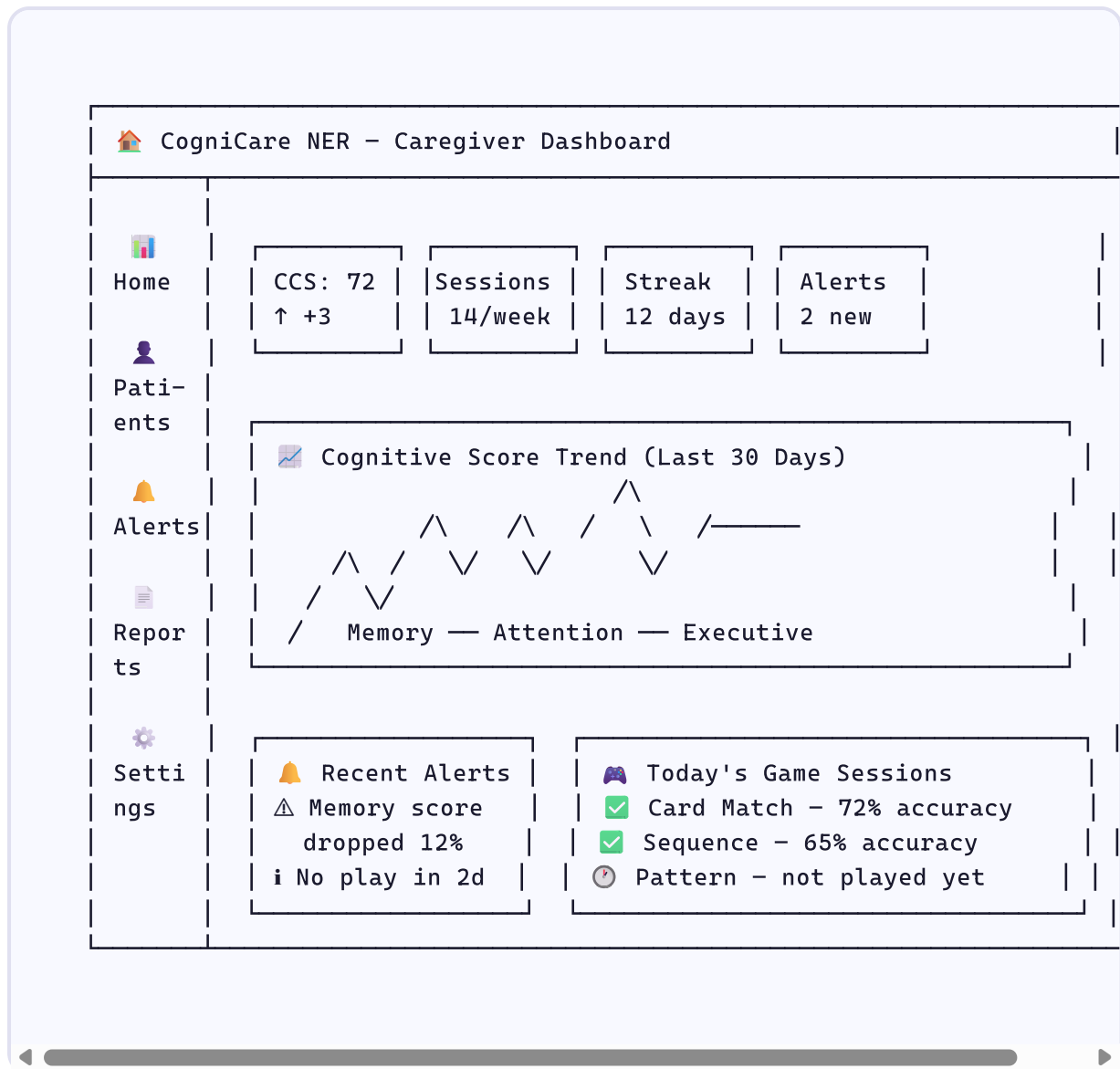
Report Generation

Downloadable PDF/CSV reports for doctor visits. Monthly cognitive progress summaries. Shareable via WhatsApp/email.

Reminder Management

Set up medicine schedules, hydration reminders, doctor appointments. Manage all from the web dashboard.

8.2 Dashboard Wireframe Concept



Core Module 5 — Reminder & Daily Assistance System

09

Gentle, persistent reminders for medication, hydration, and daily activities

9.1 Reminder Types

Type	Examples	Trigger	Escalation
 Medicine	"Time for your BP medicine"	Fixed schedule (8am, 2pm, 8pm)	Repeat every 10 min → alert caregiver after 3 misses
 Hydration	"Please drink some water"	Every 2 hours during waking hours	Gentle repeat → log compliance
 Doctor Appt	"Doctor visit tomorrow at 10am"	Day before + morning of	Alert caregiver too
 Game Time	"Let's play a brain game!"	Personalized optimal time	Suggest easiest game to start
 Exercise	"Time for a short walk"	Configurable schedule	Gentle, no pressure
 Meals	"Have you eaten lunch?"	Meal times	Alert caregiver if not confirmed

9.2 Reminder UX Design for Elderly

⚠ Critical Design Principle

Reminders for dementia patients must be **gentle, not alarming**

- . Loud alarms or flashing screens cause anxiety. Use:
- Soft chime sounds (culturally familiar tones — bamboo flute, gentle bells)
 - Warm voice narration in their language: "দোষধ খোৱাৰ সময় হ'ল" (Assamese: "Time to take medicine")
 - Large, friendly illustration on the screen (a smiling face, a water glass icon)

- Simple one-tap acknowledgment: big  button
- Never punish for missing — only gently re-nudge

Core Module 6 — Offline-First Architecture & Sync 10

Making every feature work without internet in NER's remote areas

10.1 Offline Capability Matrix

Feature	Offline	Online	Notes
All 6 Cognitive Games	✓ Full	✓ Full	Games are fully client-side
Difficulty Adaptation	✓ TF Lite	✓ Cloud ML	On-device model, updated on sync
Voice Commands	✓ Vosk	✓ Google STT	Lower accuracy offline, but functional
TTS Narration	✓ eSpeakNG	✓ Google TTS	Pre-cached common phrases
Reminders	✓ Local	✓ FCM + Local	Flutter local notifications
Game Results Storage	✓ Hive/Isar	✓ Cloud DB	Queued for sync
Cognitive Analytics	⚠ Basic	✓ Full	Basic trends on-device; full ML in cloud
Caregiver Dashboard	✗	✓ Web	Dashboard requires connectivity (caregivers likely have it)
Alerts to Caregiver	⌚ Queued	✓ Instant	Alerts queue locally, send on connectivity

10.2 Sync Protocol

SYNC PROTOCOL – CONFLICT RESOLUTION:

SYNC STRATEGY

1. DETECT CONNECTIVITY
 - └ connectivity_plus plugin checks every 30s
2. BATCH PENDING OPERATIONS
 - └ Collect all unsynced game sessions, reminders, etc.
 - └ Assign monotonic sequence numbers
3. UPLOAD (Client → Server)
 - └ POST /api/v1/sync/upload
 - └ Gzip compressed JSON payload
 - └ Server responds with accepted IDs
4. DOWNLOAD (Server → Client)
 - └ GET /api/v1/sync/download?since={last_sync_timestamp}
 - └ Updated ML model weights
 - └ Caregiver-modified reminder schedules
 - └ New cultural content packs
5. CONFLICT RESOLUTION
 - └ Game data: APPEND-ONLY (no conflicts possible)
 - └ Reminders: SERVER-WINS (caregiver is authority)
 - └ Settings: LAST-WRITE-WINS with timestamp
 - └ ML Models: SERVER-WINS (cloud has better training)
6. CONFIRM & CLEAN
 - └ Mark synced items in local queue
 - └ Prune old sync queue entries (>30 days)

Tech Stack — Complete Technology Decisions

11

Every library, framework, and service — with rationale

11.1 Mobile App (Patient-Facing)

Layer	Technology	Why
Framework	Flutter 3.x	Single codebase for Android + iOS; great offline support; smooth animations for games
State Management	flutter_bloc	Clean separation of business logic; testable; handles offline states well
Local Database	Isar 3.x	Blazing fast NoSQL for Flutter; supports complex queries; lightweight
On-Device ML	tflite_flutter	Run TensorFlow Lite models for difficulty adaptation
Offline STT	Vosk Flutter	Open-source offline speech recognition with Hindi model
Offline TTS	flutter_tts + eSpeakNG	Lightweight text-to-speech for Indic languages
Notifications	flutter_local_notifications	Reliable local scheduling; no internet needed
Network Detection	connectivity_plus	Detect online/offline state transitions
Game Rendering	Flame Engine	Lightweight 2D game engine for Flutter; perfect for card/puzzle games
Accessibility	flutter_screenutil	Responsive sizing across devices and accessibility settings

11.2 Backend Services

Layer	Technology	Why
API Framework	FastAPI (Python)	Async, fast, auto-docs (Swagger); ML ecosystem compatibility
ML Training	PyTorch + scikit-learn	Trend prediction, anomaly detection, model training
Task Queue	Celery + Redis	Background ML training, report generation, notifications
Cloud STT/TTS	Google Cloud Speech API	Best Indic language support among cloud providers
Push Notifications	Firebase Cloud Messaging	Reliable, free, cross-platform
File Storage	AWS S3 / MinIO	Game assets, audio files, cultural content packs

11.3 Caregiver Dashboard

Layer	Technology	Why
Framework	Next.js 14 (React)	SSR for fast loads; API routes; great UX ecosystem
Charts	Recharts + Chart.js	Beautiful, responsive charts for cognitive data visualization
Styling	Tailwind CSS	Rapid development; responsive design; clean UI
State	Zustand	Lightweight, simple state management

11.4 Database & Infrastructure

Component	Technology	Why
Primary DB	PostgreSQL 16	Users, auth, relational data — battle-tested reliability
Analytics DB	MongoDB	Game sessions, performance metrics — flexible schema for evolving analytics
Cache	Redis	Session cache, real-time data, task queue backend
Container	Docker + Docker Compose	Consistent dev/prod environments; easy SIH demo setup
Hosting	DigitalOcean / AWS Lightsail	Cost-effective for hackathon stage; easy to scale later

Database Design & Data Models

Schema design for relational and document databases

12

12.1 PostgreSQL Schema (Core Entities)

```
-- Users & Authentication
CREATE TABLE users (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  phone VARCHAR(15) UNIQUE NOT NULL,
  role VARCHAR(20) NOT NULL CHECK (role IN ('patient', 'caregiver', 'healthcare_provider')),
  name VARCHAR(100) NOT NULL,
  preferred_language VARCHAR(10) DEFAULT 'hi',
  created_at TIMESTAMPTZ DEFAULT NOW(),
  is_active BOOLEAN DEFAULT TRUE
);

-- Patient Profiles
CREATE TABLE patients (
  id UUID PRIMARY KEY REFERENCES users(id),
  date_of_birth DATE NOT NULL,
  gender VARCHAR(10),
  state VARCHAR(50), -- NER state
  district VARCHAR(100),
  dementia_stage VARCHAR(20) CHECK (dementia_stage IN ('mild', 'moderate', 'severe')),
  diagnosis_date DATE,
  primary_caregiver_id UUID REFERENCES users(id),
  emergency_contact VARCHAR(15),
  medical_notes TEXT,
  cognitive_baseline_score FLOAT
);

-- Caregiver-Patient Relationships
CREATE TABLE caregiver_patient (
  caregiver_id UUID REFERENCES users(id),
  patient_id UUID REFERENCES patients(id),
  relationship VARCHAR(50), -- 'family', 'nurse', 'doctor'
  assigned_at TIMESTAMPTZ DEFAULT NOW(),
  PRIMARY KEY (caregiver_id, patient_id)
);

-- Reminder Schedules
CREATE TABLE reminders (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```

    patient_id UUID REFERENCES patients(id),
    type VARCHAR(20) NOT NULL, -- 'medicine', 'hydration', 'appointment',
    title VARCHAR(200) NOT NULL,
    description TEXT,
    schedule_cron VARCHAR(100), -- cron expression for recurring
    one_time_at TIMESTAMPTZ,    -- for one-time reminders
    voice_message_url VARCHAR(500),
    is_active BOOLEAN DEFAULT TRUE,
    created_by UUID REFERENCES users(id)
);

```

12.2 MongoDB Schema (Game Analytics)

```

// Game Session Document
{
  "_id": ObjectId,
  "patient_id": "uuid-string",
  "game_type": "card_match",      // card_match | sequence_recall | spot_d
  "difficulty_level": 3,          // 1-5
  "started_at": ISODate,
  "ended_at": ISODate,
  "duration_seconds": 180,
  "metrics": {
    "accuracy": 0.72,
    "avg_response_time_ms": 2340,
    "response_time_variance": 450,
    "total_attempts": 12,
    "correct_attempts": 9,
    "hints_used": 1,
    "pauses": 2,
    "error_types": ["near_miss", "random_guess"],
    "touch_precision_score": 0.85
  },
  "cognitive_scores": {
    "memory": 68,
    "attention": 75,
    "executive": 70,
    "language": null,             // not tested in this game
    "visuospatial": 72
  },
  "ai_decisions": {
    "difficulty_before": 3,
    "difficulty_after": 3,
    "fatigue_detected": false,
    "flow_zone_achieved": true,
    "next_recommendation": "sequence_recall"
  }
}

```

```
},  
  "device_info": {  
    "was_offline": true,  
    "synced_at": ISODate  
  }  
}
```



UI/UX Design — Elderly-Friendly Interface

13

Designing for users aged 65+ with cognitive impairments

13.1 Design Principles for Elderly Dementia Users

Principle	Implementation	Rationale
Large Touch Targets	Minimum 60×60dp for all interactive elements	Reduced motor precision due to aging + tremors
High Contrast	WCAG AAA (7:1 ratio); dark text on light backgrounds	Age-related vision decline; cataracts common
Large Text	Base font 20sp; headers 28sp+; support system scaling	Presbyopia affects nearly all 65+ adults
Simple Navigation	Max 2 levels deep; persistent bottom bar with 3-4 items	Reduced working memory; complex nav causes confusion
Consistent Layout	Same button positions, same color meanings across all screens	Familiarity reduces cognitive load
No Abstract Icons	All icons paired with text labels; use realistic illustrations	Abstract symbols may be unrecognizable
Warm Colors	Earth tones, warm pastels; avoid cold blues and harsh reds	Warm colors feel safe and calming; reduce anxiety
Gentle Feedback	Soft animations, happy sounds for success; no harsh failure sounds	Prevent frustration and anxiety
Voice Everything	Every screen reads aloud; voice input for everything	Bypass reading/typing difficulties
No Time Pressure	Timers are suggestions, not hard limits (except where needed for testing)	Time pressure causes anxiety and worsens performance

13.2 Color Palette

🌿 Primary — Warm Earth

#5D4037 (Bark Brown)

#FFF8F0 (Cream)

Warm, familiar, calming

✅ Success — Gentle Green

#2E7D32 (Forest Green)

#E8F5E9 (Mint)

Positive, encouraging

🔔 Attention — Warm Orange

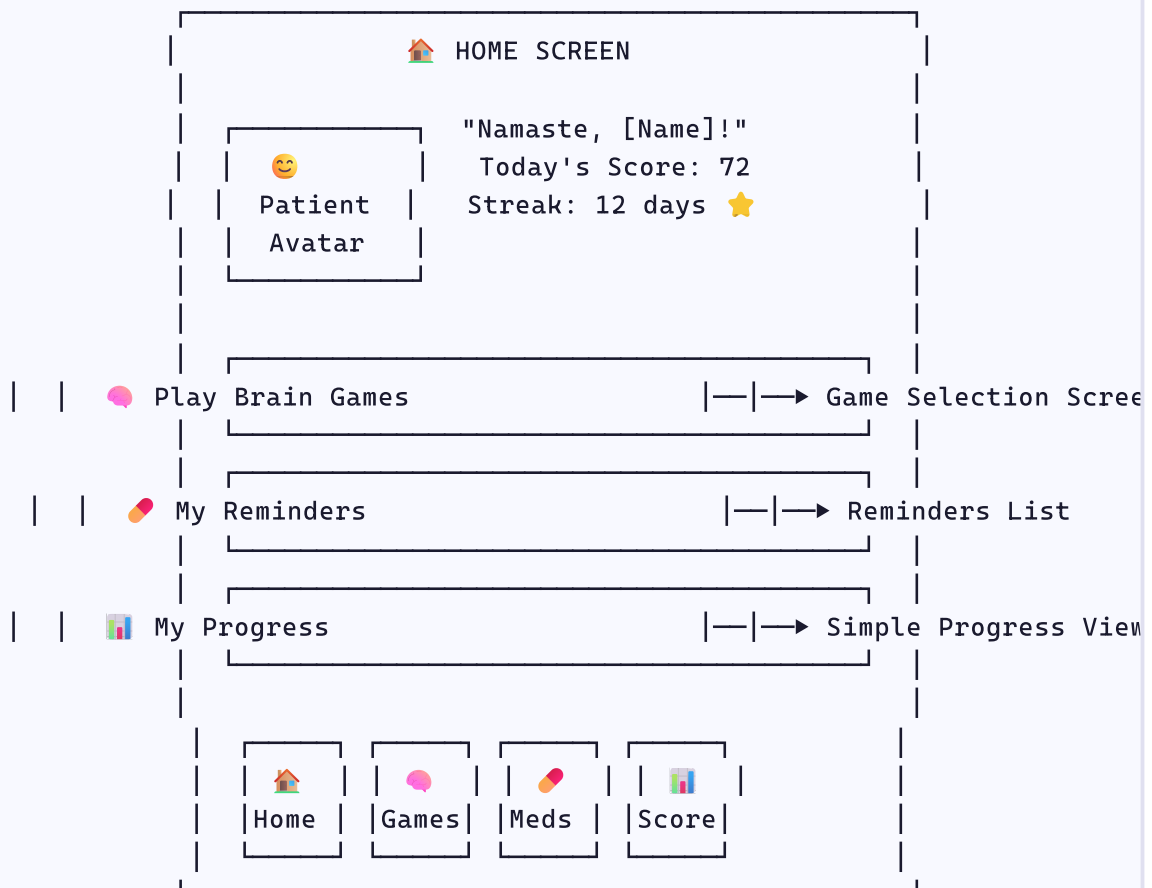
#E65100 (Deep Orange)

#FFF3E0 (Peach)

Gentle attention-getting

13.3 App Screen Flow

APP NAVIGATION FLOW:



All screens have:

- ✅ Large 60dp+ touch targets
- ✅ Voice narration on entry
- ✅ Back button always visible
- ✅ No more than 4 choices per screen

Security, Privacy & Compliance

14

Protecting sensitive health data of vulnerable elderly patients

14.1 Data Protection Measures

Layer	Measure	Implementation
Data at Rest	AES-256 encryption	Encrypted Isar DB on device; encrypted PostgreSQL columns for PII
Data in Transit	TLS 1.3	HTTPS for all API calls; certificate pinning in Flutter
Authentication	JWT + Refresh Tokens	Short-lived access tokens (15 min); secure refresh token rotation
Authorization	Role-Based Access Control	Patient, Caregiver, Healthcare Worker, Admin roles with strict permissions
Anonymization	Pseudonymized analytics	Game performance data stripped of PII for aggregate analysis
Audit Logging	All access logged	Who accessed what patient data and when — tamper-proof logs
Consent	Explicit consent flow	Caregiver provides informed consent for data collection on behalf of patient

14.2 Regulatory Compliance

- **Digital Personal Data Protection Act (DPDPA) 2023** — India's data protection law; comply with data minimization, purpose limitation, consent requirements
- **IT Act 2000 (Section 43A)** — Reasonable security practices for sensitive personal data
- **HIPAA principles** — Though not legally required in India, following HIPAA-like practices (minimum necessary, access controls, audit trails) strengthens the solution
- **Accessibility Guidelines** — WCAG 2.1 Level AA compliance for the caregiver dashboard

Execution Roadmap — Week-by-Week Plan

15

A detailed 12-week development plan from now to SIH Grand Finale



Roadmap Assumptions

Deadline: 30 September 2026 (SIH idea submission). Assuming a 6-member team with ~3 weeks remaining and potential Grand Finale in December 2026. This roadmap covers: (1) Idea Phase → (2) Prototype → (3) Full MVP → (4) Grand Finale Prep.

Phase 1: Research & Foundation (Week 1-2)



Week 1 — Research & Design

Deep Domain Research + Architecture Design

- Research dementia cognitive training literature — Lampit et al., Sea Hero Quest study
- Study NER demographics, languages, cultural elements (ARDSI Dementia India Report)
- Finalize system architecture and tech stack
- Create detailed wireframes and user flows in Figma
- Set up Git repository, CI/CD pipeline, project management (Jira/Linear)
- Write the SIH idea submission document

Design

Research



Week 2 — Foundation Setup

Project scaffolding + Core infrastructure

- Set up Flutter project with folder structure and navigation
- Set up FastAPI backend with project structure, authentication
- Design and create PostgreSQL + MongoDB schemas
- Implement basic auth flow (phone OTP login)
- Set up Docker Compose for local development
- Integrate Isar DB on mobile for offline storage

Flutter

FastAPI

Database

Phase 2: Core Development (Week 3-6)

Week 3 — Game Engine MVP

First 2 games playable with static difficulty

- Build Card Match game with NER cultural card designs
- Build Sequence Recall game with cultural themes
- Implement game session data collection framework
- Create reusable game UI components (timer, score display, progress bar)
- Design and integrate 20+ NER cultural illustrations

Flutter

Design

Week 4 — More Games + Backend APIs

4 games complete + backend APIs operational

- Build Spot the Difference and Pattern Completion games
- Implement game session API endpoints (submit, history)
- Build patient profile management APIs
- Implement reminder CRUD APIs and scheduling service
- Set up Celery task queue for background processing

Flutter

FastAPI

Week 5 — AI/ML Integration

Difficulty adaptation working end-to-end

- Implement Thompson Sampling for difficulty adaptation
- Train initial TensorFlow Lite model and integrate on-device
- Build CCS (Composite Cognitive Score) calculation engine
- Implement decline detection algorithms
- Build remaining 2 games (Word Association, Daily Routine Sort)

AI/ML

Flutter

Week 6 — Voice & Language

Multilingual voice interaction operational

- Integrate Google Cloud Speech API for online STT

- Integrate Vosk for offline STT (Hindi model)
- Implement voice command intent classification
- Add TTS narration for all screens and game instructions
- Build language selection and localization framework (i18n)
- Add Assamese, Bengali, Hindi content packs

NLP

Flutter

Phase 3: Advanced Features (Week 7-9)

Week 7 — Caregiver Dashboard

Web dashboard with patient tracking

- Build Next.js dashboard with authentication
- Implement cognitive trend charts (Recharts)
- Build alert center with real-time updates (WebSocket)
- Create patient profile views with history
- Implement report generation (PDF export)

Next.js

Dashboard

Week 8 — Offline & Sync

Full offline capability + sync engine

- Implement offline queue system for game results
- Build bidirectional sync protocol
- Implement conflict resolution logic
- Test offline → online transitions extensively
- Implement ML model weight downloading via sync
- Build local-only analytics for offline cognitive scoring

Sync

Backend

Week 9 — Reminders + Alerts

Complete reminder system + caregiver alerts

- Implement flutter_local_notifications for all reminder types
- Build caregiver alert system (push notifications via FCM)

- Create gentle UI for reminder presentations (voice + visual)
- Implement reminder compliance tracking
- Build escalation logic (missed medicine → alert caregiver)

Flutter

Backend

Phase 4: Polish & Grand Finale Prep (Week 10-12)

Week 10 — Testing & Refinement

Comprehensive testing + UX polish

- Unit tests for all backend services (pytest, 80%+ coverage)
- Widget tests for Flutter games and UI
- Integration tests for sync engine
- Accessibility audit (screen reader, font scaling, contrast)
- Performance optimization (load times, animation smoothness)
- User testing with 2-3 elderly volunteers (if possible)

Testing

Polish

Week 11 — Demo & Documentation

Demo video + comprehensive documentation

- Record 3-5 min demo video showing full user journey
- Write technical documentation (API docs, architecture diagrams)
- Create PowerPoint/Google Slides presentation (20-25 slides)
- Prepare for Q&A — anticipate judge questions
- Deploy backend to cloud (DigitalOcean/AWS)
- Publish APK build for testing

Demo

Deploy

Week 12 — Grand Finale Preparation

Final rehearsals + backup plans

- Practice presentation 5+ times with mock Q&A
- Ensure demo works on airplane mode (offline demo)
- Prepare backup: screen recordings in case of live demo failure

- Final bug fixes and edge case handling
- Team member role assignments for presentation
- Pack hardware: laptop, phone, chargers, hotspot

Prep

Team Structure & Role Allocation

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Optimal 6-member team composition for SIH

Role	Responsibilities	Key Skills Needed
 Team Lead / PM	Overall coordination, timeline management, SIH paperwork, presentation lead speaker	Communication, project management, public speaking
 Flutter Developer	Patient mobile app — games, UI, offline storage, voice integration	Flutter/Dart, Flame engine, state management (Bloc), Isar
 AI/ML Engineer	Difficulty adaptation algorithm, cognitive scoring, trend prediction, TF Lite model	Python, PyTorch, TensorFlow Lite, scikit-learn, reinforcement learning
 Backend Developer	FastAPI services, database design, sync engine, authentication, APIs	Python, FastAPI, PostgreSQL, MongoDB, Redis, Docker
 UI/UX Designer + Frontend Dev	Figma designs, NER cultural illustrations, caregiver dashboard (React/Next.js)	Figma, React/Next.js, Tailwind, accessibility design, illustration
 Research + Testing	Domain research, testing, documentation, data collection, voice/language integration	Research skills, testing, documentation, NLP basics, NER cultural knowledge

SIH Evaluation Criteria & Scoring Strategy

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How to maximize your scores on every criterion

Criteria	Weight	What Judges Look For	How to Score Maximum
Novelty / Innovation	~20%	Is this genuinely new? Not just another health app?	Emphasize: NER cultural AI gaming (first of its kind), Thompson Sampling adaptation (not simple threshold), offline ML inference, composite cognitive scoring
Complexity / Technical Depth	~20%	Is the engineering real? Can they explain the ML?	Show: Architecture diagrams, working ML model with real difficulty changes, offline sync working live, database schemas. No hand-waving
Feasibility / Practicality	~20%	Can this actually be built and deployed in NER?	Demo offline mode, show low-bandwidth sync, mention partnering with LGBRIMH Tezpur, discuss deployment via ASHA workers
Social Impact	~15%	Will this meaningfully help elderly dementia patients?	Cite cognitive science research, show potential for early intervention, discuss quality-of-life improvement, mention SDG 3 alignment
User Experience	~15%	Is the UI actually usable by a 70-year-old with dementia?	Live demo with large text, voice narration in Assamese, gentle animations, one-tap interactions. Show you tested with elderly UX principles
Presentation / Communication	~10%	Can the team explain their solution clearly?	Structured presentation, clear demo flow, confident Q&A, team coordination. Start with the problem (emotional NER elderly story), end with impact

Presentation & Demo Strategy

How to present at the SIH Grand Finale for maximum impact

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18.1 Presentation Structure (7 minutes)

Time	Content	Speaker	Tip
0:00 – 0:45	Hook: "Meet Lakhimi, 72, from a village in Assam..." (tell a story)	Team Lead	Make it emotional. Show a photo of NER elderly life.
0:45 – 1:30	Problem: Dementia stats, NER challenges, gap in existing solutions	Team Lead	Use numbers: 5M+ dementia patients, 200+ NER languages, 0 culturally relevant solutions
1:30 – 3:00	Solution Demo: Live walkthrough — open app, play a game, show voice, show offline	Flutter Dev	Keep phone screen mirrored. Show Assamese voice narration. Go to airplane mode for offline demo
3:00 – 4:00	AI/ML Deep Dive: How difficulty adapts, show CCS trend, decline detection	ML Engineer	Show actual data: "Patient played 14 sessions, CCS went from 55 to 68, difficulty adapted from 2 to 3"
4:00 – 5:00	Caregiver Dashboard: Show web dashboard with charts, alerts, reports	UI/Frontend Dev	Show the dashboard with realistic data. Click through alerts.
5:00 – 5:45	Architecture & Tech: Quick architecture slide, offline sync, security	Backend Dev	One clean architecture diagram. Don't over-explain — show depth through Q&A
5:45 – 6:30	Impact & Future: Potential reach, scalability to all of India, partnership with ARDSI	Team Lead	End with Lakhimi's story again: "With CogniCare, Lakhimi plays games in Assamese every morning..."
6:30 – 7:00	Close: Call to action, thank you	Team Lead	Confident close. "We're ready to deploy."

18.2 Anticipated Judge Questions & Answers

Question	Strong Answer
"How do you validate that the games actually help dementia patients?"	"We base our game design on peer-reviewed research — Lampit et al.'s meta-analysis shows computerized cognitive training improves cognition in older adults. We plan to validate with a pilot at LGBRIMH Tezpur, Assam, tracking CCS over 3 months vs a control group."
"How does the AI difficulty adaptation work technically?"	"We use Thompson Sampling — a contextual multi-armed bandit — targeting the 60-80% accuracy flow zone. On-device TF Lite model (~500KB) runs inference. Cloud model trains on aggregated data and pushes updated weights via sync."
"What about data privacy for vulnerable elderly patients?"	"We follow DPDPA 2023 guidelines. Data encrypted at rest (AES-256) and transit (TLS 1.3). Caregiver provides informed consent. Analytics are pseudonymized. RBAC ensures only authorized caregivers access patient data."
"How does it work without internet?"	"Core features — all 6 games, difficulty adaptation via TF Lite, reminders via local notifications, voice via Vosk — work 100% offline. Data queues locally in Isar DB and syncs when connectivity returns. We live-demoed this on airplane mode."
"Why Flutter and not native Android?"	"Flutter gives us cross-platform from one codebase, smooth game animations via the Flame engine, and excellent offline DB options (Isar). For a hackathon team, it maximizes what we can build in the time we have."
"How will you scale this beyond the hackathon?"	"Phase 1: Pilot in Assam with LGBRIMH. Phase 2: Expand language support + partner with NHM (National Health Mission) for NER deployment. Phase 3: Adapt for pan-India with regional cultural packs. The architecture is modular — adding a new language or cultural pack is configuration, not code."

Innovation & USP Differentiators

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What makes CogniCare NER stand out from any other team

19.1 Unique Selling Points

#	USP	Why It's Novel
1	NER Cultural Intelligence	No existing cognitive game platform incorporates North Eastern Indian cultural elements — Gamosa patterns, Bihu steps, Naga shawls, bamboo dance formations. This makes games feel familiar, not foreign, to elderly NER users.
2	Thompson Sampling AI Adaptation	Most brain training apps use simple threshold rules ("if score > 80, increase difficulty"). Thompson Sampling handles the exploration-exploitation tradeoff — the system discovers each patient's optimal challenge level, not just reacts to their last score.
3	Composite Cognitive Score (CCS)	A unified cognitive metric across 5 domains, computed from gameplay data — not a medical test. Provides continuous, non-invasive cognitive monitoring that can supplement annual clinical assessments.
4	Offline-First with On-Device ML	The entire AI pipeline (difficulty adaptation, fatigue detection, game recommendation) runs on-device via TF Lite. No "this feature requires internet" compromise.
5	NER Language Voice Interface	Voice commands and TTS in Assamese, Manipuri, Mizo — languages no commercial health app supports. Offline voice via Vosk means it works in areas with no connectivity.
6	Caregiver-in-the-Loop Design	Most health apps treat caregivers as an afterthought. CogniCare puts them at the center — dashboards, alerts, remote reminder management, report generation for doctor visits.

19.2 Potential Future Extensions

- **AR-based reminiscence therapy** — Show old NER photos/places in AR to trigger memories
- **Wearable integration** — Connect with basic fitness bands for sleep/activity correlation with cognitive performance

- **Telemedicine bridge** — Let rural NER patients video-call neurologists using the app's cognitive data as context
- **Community health worker (ASHA) integration** — ASHA workers can onboard elderly patients and monitor via the dashboard
- **Multi-player cognitive games** — Elderly can play with grandchildren remotely, improving engagement and social connection
- **Longitudinal data for research** — Anonymized aggregate data could contribute to the world's largest NER-specific dementia dataset

Research References & Resources

Key papers, datasets, tools, and resources for the team

20

20.1 Key Research Papers

1. **Lampit, A., Hallock, H., & Valenzuela, M. (2014).** "Computerized Cognitive Training in Cognitively Healthy Older Adults: A Systematic Review and Meta-Analysis of Effect Modifiers." *PLOS Medicine*. — Foundational evidence for computerized cognitive training efficacy.
2. **Coughlan, G., et al. (2019).** "Toward Personalized Cognitive Diagnostics of At-Genetic-Risk Alzheimer's Disease." *PNAS*. — Sea Hero Quest data for spatial navigation and dementia detection.
3. **Manera, V., et al. (2015).** "An ICT-Based Approach for Cognitive Stimulation in Older Adults." *Frontiers in Aging Neuroscience*. — Serious games for cognitive stimulation.
4. **ARDSI (2020).** "Dementia India Report 2020." *Alzheimer's and Related Disorders Society of India*. — India-specific dementia epidemiology.
5. **Zohaib Hassan M., et al. (2023).** "Deep Reinforcement Learning for Personalized Cognitive Rehabilitation." — AI techniques for adaptive difficulty.

20.2 Open-Source Tools & Libraries

Tool	Purpose	Link
Flutter	Mobile app framework	flutter.dev
Flame Engine	2D game engine for Flutter	flame-engine.org
Isar	Local offline database	isar.dev
FastAPI	Python backend framework	fastapi.tiangolo.com
Vosk	Offline speech recognition	alphacephei.com/vosk
IndicTTS	TTS for Indian languages	iitm.ac.in/donlab/tts
TensorFlow Lite	On-device ML inference	tensorflow.org/lite
Recharts	React charting library	recharts.org

20.3 Key Organizations to Reference / Partner With

- **ARDSI** (Alzheimer's and Related Disorders Society of India) — ardsi.org — India's primary dementia organization
- **LGBRIMH** (Lokopriya Gopinath Bordoloi Regional Institute of Mental Health, Tezpur, Assam) — Leading mental health institution in NER
- **MDoNER** (Ministry of Development of North Eastern Region) — The problem statement sponsor
- **NHM** (National Health Mission) — Government health program with ASHA worker infrastructure in NER
- **IIT Guwahati** — NER's premier tech institute, potential research partner

20.4 Useful Datasets

- **OASIS (Open Access Series of Imaging Studies)** — MRI brain scans for dementia research
- **Sea Hero Quest Open Dataset** — Spatial navigation norms for dementia detection
- **TIMIT / CommonVoice** — Speech recognition training data (adapt for NER languages)
- **IndicNLP Catalog** — NLP resources for Indian languages (useful for NER language models)

Final Advice

The teams that win SIH are the ones that show
depth over breadth

. Don't try to build everything — pick 2-3 games with beautiful cultural content, nail the AI adaptation, make offline work flawlessly, and rehearse your demo until it's bulletproof. A polished 60% is better than a broken 100%.

Good luck! 

CogniCare NER — SIH26003 Complete Guide

Prepared for Smart India Hackathon 2026 | Generated September 2026

This guide is for educational and hackathon preparation purposes.